# A3-PS014-TC169

## Pilot Internet-Quality Comparison by Verified Access Medium

**Category:** Smart Cities
**Presented by:** SYNORA
**Team/Problem ID:** A3-PS014-TC169
**Event:** AIORI Hackathon 3

# Problem Description

**Limitations of Conventional Speed Tests**

Traditional speed tests mainly focus on download and upload speeds, so they may not provide a complete picture of overall internet quality.

**Variation in Internet Quality**

Network performance can vary depending on the access medium, ISP, location, time of day, and whether the device uses Wi-Fi or Ethernet. Poor Wi-Fi quality may also be mistaken for an issue with the internet connection itself.

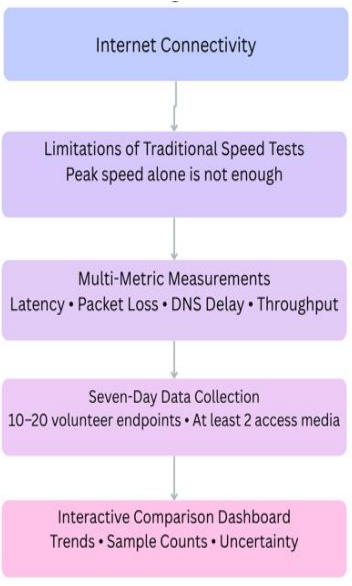
**Impact of Network Performance**

Latency, packet loss, and DNS resolution delays can affect website loading, video calls, online gaming, and other real-time applications.

**Need for a Reliable Comparison Platform**

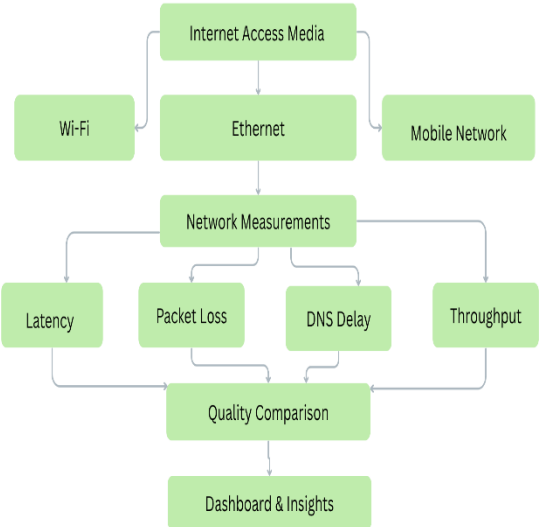
A controlled and repeatable measurement platform is needed to collect multiple network performance metrics over time and compare internet quality fairly across different access media and network conditions.

## Internet Quality Measurement



# Background

- Internet quality depends on more than download speed; reliability and response time also matter.
- Different access media, such as Wi-Fi, Ethernet, and mobile networks, can deliver different performance.
- Latency, packet loss, DNS delay, and throughput help evaluate network quality.
- Performance can fluctuate with network congestion, signal strength, and time of day
- Standardized measurements help make fairer comparisons between access media.



# Literature & RFC Review

**01 — Measurement Framework**
RFC 2330
What it gives us: A framework for defining network performance metrics and measurement methods.
OUR TAKEAWAY → Make measurements repeatable.

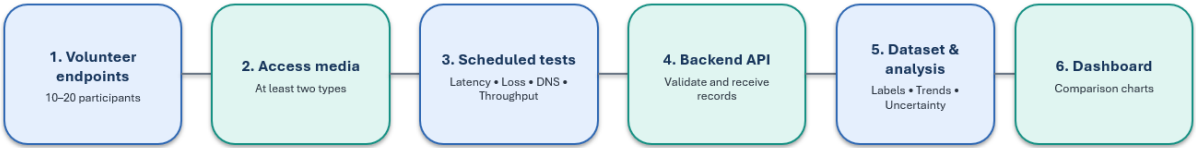
**02 — Conventional Speed Tests**
Useful, but incomplete
What they can miss: Packet loss, DNS delay, and performance changes over time.
OUR TAKEAWAY → Speed alone is not enough.

**03 — Multi-Metric Assessment**
A broader view of network quality
What we measure: Latency · Packet loss · DNS delay · Throughput
OUR TAKEAWAY → Measure repeatedly and compare trends.

**04 — The Gap We Address**
From isolated tests to fair comparisons
Our contribution: Compare access media using repeated measurements, labelled connection details, sample counts, and uncertainty.
OUR TAKEAWAY → Explain differences, not just speeds.

# Proposed Solution

End-to-end seven-day pilot measurement workflow




Outputs: access-medium comparisons, time-series trends, sample counts, and limitations.

# Optimization Proposed

| Aspect | Before | After: Our Optimization |
|---|---|---|
| Metrics | Mainly speed | Latency, loss, DNS, throughput |
| Schedule | Single test | Scheduled for 7 days |
| Connection | Limited context | Label access medium and Wi-Fi/Ethernet |
| est setup | Conditions may differ | Controlled endpoint |
| Analysis | Individual values | Trends and participant summaries |
| Reliability | Limited statistical context | Sample counts, variation, uncertainty |
| Conclusions | Risk of broad claims | Pilot-specific findings and limitations |

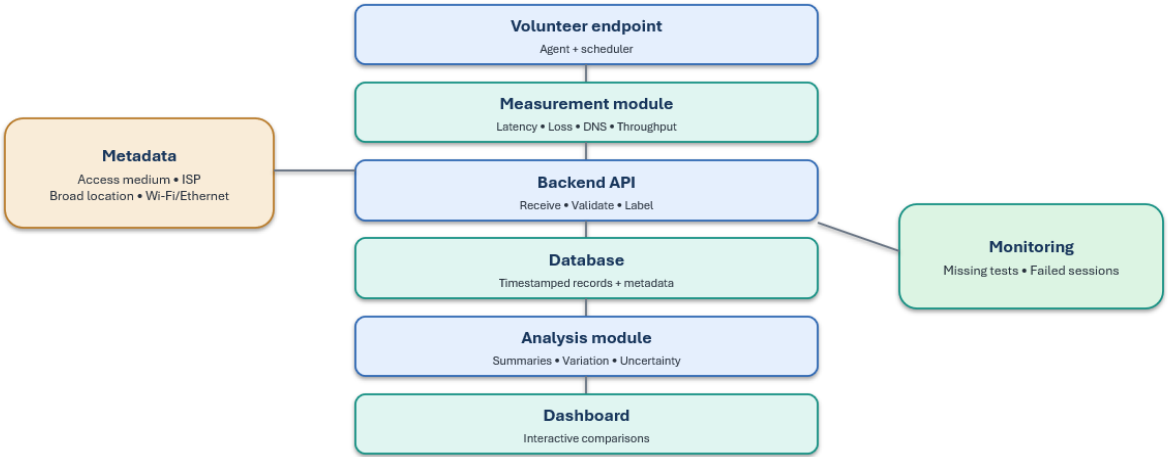
**Standardize**
Consistent probes, targets and timeouts.

**Control traffic**
Limit throughput-test size and frequency.

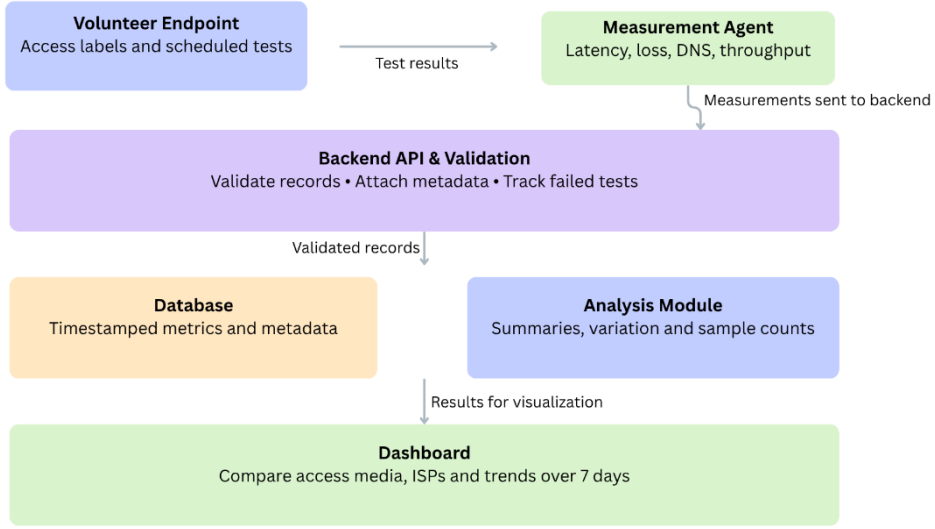
**Separate causes**
Compare gateway and end-to-end results where feasible.

**Preserve evidence**
Keep raw observations; distinguish missing data from failures.

# Solution Architecture & Design



# Data Flow Diagram



# Deployment Timeline

The project requires seven days of scheduled measurements. Your implementation and participant onboarding should therefore be planned around the data-collection window.

The following is a proposed development timeline. Adjust the calendar dates to your actual hackathon deadline.

| Phase | Duration | Activities and milestone |
|---|---|---|
| Phase 1 | Days 1–2 | Define metrics, architecture, database schema, and measurement protocol |
| Phase 2 | Days 3–4 | Develop the measurement agent and controlled endpoint |
| Phase 3 | Days 5–6 | Implement the backend, data storage, and basic dashboard |
| Phase 4 | Day 7 | Test the full pipeline and onboard 10–20 volunteers |
| Phase 5 | Days 8–14 | Conduct seven days of scheduled measurement collection |
| Phase 6 | Days 15–16 | Validate the dataset and analyze trends and uncertainty |
| Phase 7 | Days 17–18 | Finalize the dashboard, report, presentation, and demonstration |

# Bibliography


1. Paxson, V., Almes, G., Mahdavi, J., & Mathis, M. (1998). *Framework for IP Performance Metrics (RFC 2330)*. IETF. https://www.rfc-editor.org/rfc/rfc2330.html
2. AIORI-3 Hackathon. *PS-014: Pilot Internet-Quality Comparison by Verified Access Medium.* Problem Statement, Smart Cities Category.
3. AIORI-3 Hackathon. *Proposed Structure for Hackathon Presentation.* Reference Presentation Template, October 2026.

# THANK YOU

Pilot Internet-Quality Comparison by verified Access Medium

Smarter connectivity for smarter cities